CE F230
Civil Engineering Materials
Concrete

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Concrete Supports Modern Lifestyle

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Concrete Applications

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- ❖ Residential and commercial buildings
- ❖ Bridges, flyovers, culverts
- ❖ Dams, tunnels, water tanks
- ❖ Swimming pools
- ❖ Roads, runways, pipes
- ❖ Foundations, piles, sewers
- ❖ Offshore platforms
- ❖ Nuclear power stations, radiation shields etc.
- ❖ Fire and corrosion protection of steel structures
- ❖ Many more applications…

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Concrete Facts

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- ❖ The most used man-made material
- ❖ The 2nd most used material
- ❖ Total value of concrete infrastructure > 17 trillion US dollars
- ❖ Annual consumption of concrete in the world
  - 18 billion ton/year (as of 2006)
  - About 3 tons per person
  - More than 10x that of steel
- ❖ Why is concrete widely used as a construction material?

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Concrete as a Construction Material- Advantages

- Ease of production from local materials and experience (cost benefit)
- Mouldability to achieve any shape and size
- A durable material in principle
- Excellent material for fire resistance
- Requires less energy to produce than other construction materials
- Aesthetic possibilities through the use of color, texture, and shape
- A material with tailorable properties

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Energy Consumption for Production of Several Construction Materials

Material	Energy Requirement (GJ/m^3)
Aluminum	360
Steel	300
Glass	50
Concrete	3.4

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What is Concrete?

- Artificial stone produced from sand and stone together with cement paste (cement + water)
- Cement paste fills the space between stone and sand particles
- Concretus (a Latin word)
  - Compact
  - Condensed
  - To grow together

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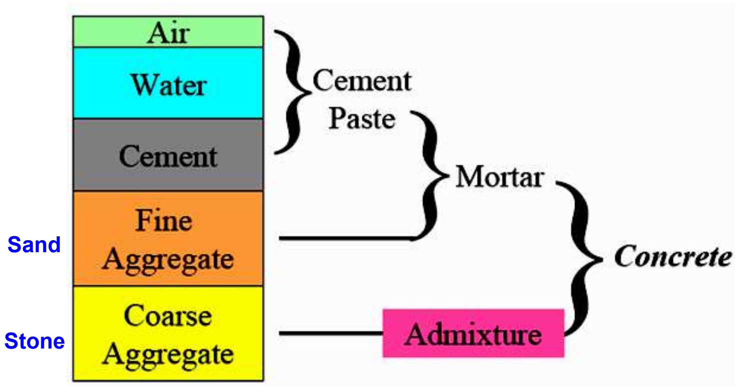
Paste, Mortar, or Concrete?



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Ingredients of Modern Concrete

- Binder materials
  - Portland cements (various types)
  - Supplementary cementitious materials (SCM)
- Aggregates
  - Fine aggregates < 5mm
  - Coarse aggregates > 5mm
- Water
- Admixtures
- Fiber

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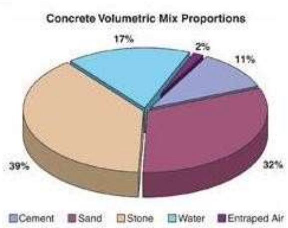
Proportions of Ingredients



Volumetric composition of concrete

- Cement or binder = 6 to 16 %
- Water = 12 to 20 %
- Fine aggregates = 20 to 30 %
- Coarse aggregates = 40 to 55 %
- Air content = 1 to 3 % (non air-entrained)
  = 4 to 8 % (air-entrained)

Optional components

- Admixture
- Supplementary cementitious materials
  - Silica fume
  - Fly ash
  - Slag
  - Pozzolans
- Fibers
  - Steel
  - Polypropylene
  - Glass
  - Carbon
  - Nylon
  - Natural etc.

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Typical Properties of Structural Concrete

Compressive strength	35 MPa
Flexural strength	6 MPa
Tensile strength	3 MPa
Modulus of elasticity	28 GPa
Poisson's ratio	0.18
Tensile strain at failure	0.001
Coefficient of thermal expansion	10 x 10^{-6}/°C
Ultimate shrinkage strain	0.05-0.1%
Density	
Normal weight	2300 kg/m^3
Lightweight	1800 kg/m^3

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Concrete as a Construction Material - Disadvantages

- A brittle material with very low tensile strength and tensile ductility.
- Even in compression, concrete has a relatively low strength-to-weight ratio
- Deformations: Shrinkage and creep

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Advantages and Disadvantages of Concrete as a Construction Material

Advantage	Disadvantage
Ability to be cast	Low tensile strength
Economical	Low ductility
Durable	Volume instability
Fire resistant	Low strength-to-weight ratio
Energy efficient	
On-site fabrication	
Aesthetic properties	

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Environmental Impact of Concrete

- Enormous raw material and energy consumption
  - Global concrete demand > 18 billion tons annually as of 2006
  - 1 ton of cement clinker requires 1.7 tons of non-fuel raw materials
  - Cement production is 10x more energy intensive than general economy and is account for 2% of global primary energy use (4000-7500 MJ per tonne of cement)
  - Land scarring
- CO_2 emission and climate change
  - Production of 1 ton of cement clinker generates equal amount of greenhouse gas
  - Cement production accounts for 5-10% of global CO_2 emissions
- Surface runoff
- Urban heat island

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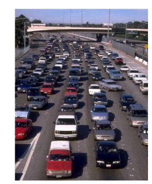
Great Challenge for Civil Engineers: Co-existence Between the Natural and Built Environment

Greenhouse gas emissions



Energy consumption

Or a battle?

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What You Will Learn About Concrete in this Course …

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Cementitious Materials and their Hydration

Hydrated cement paste

Photo courtesy of Mr. Jim Margeson, NRC-IRC

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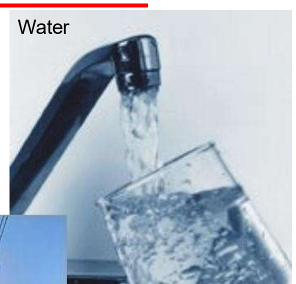
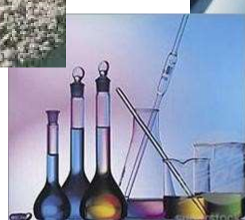
Properties of Other Concrete Ingredients

Water





Admixtures

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Mix Design: How to Design A Concrete Mix w.r.t. A Specified Compressive Strength?

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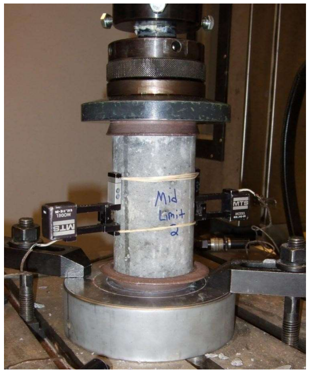
Fresh and Hardened Properties of Concrete and Tests

Slump test on fresh concrete

Compression test on hardened concrete



Concrete Durability

Deterioration due to alkali-aggregate reaction

Deterioration due to corrosion of steel reinforcing bars